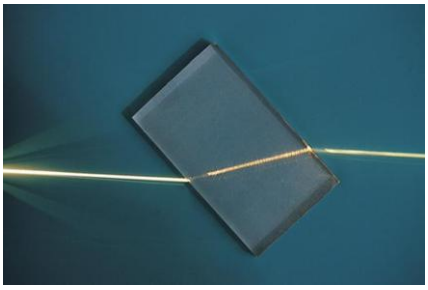
Chapter 58, Refraction of Light
Calculus Physics II

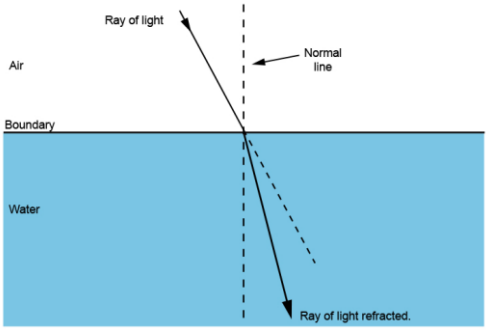
Refraction of EM waves - Refraction of EM waves is due to the transition of the wave from one material to another with a change in speed of the wave at the boundary of the two materials.





Interface – The boundary between two materials or medium.

Normal line – A line that is drawn perpendicular to the interface and all angles are measured relative to that line

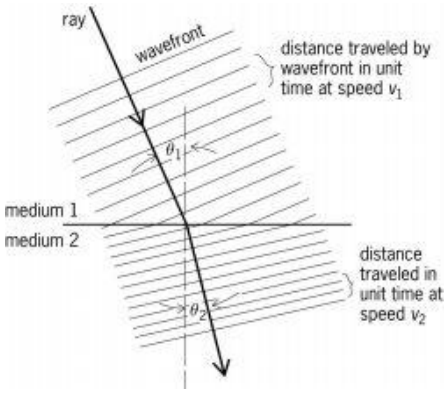


Refraction: What happens when EM waves pass through a material?

At the interface of two different and transparent materials, light is transmitted and reflected, for example let us use the air/glass interface of a window.

1) As the wave goes from material 1and enters the second material (glass) the light changes speed (slows down for air to glass). By slowing down the wave in the material does not go as far in the same amount of time as the wave outside the material. The only way the two waves (inside and outside the material) can stay connected is for the wave in the material to change direction.

2) The rays of light bend towards the Normal to the interface because the wave that is in the material slows down and travels a shorter distance in one cycle than the wave outside the medium. The change of speed (slowing down) changes the wavelength of the EM wave; this means the wave in the material has a shorter wavelength.



3) The frequency of the waves does not change, no matter what material the light is travelling through (the air or the glass). The frequency of the wave is set by what created the wave. The velocity changes and this creates a change in the wavelength of the light going from one medium to another.

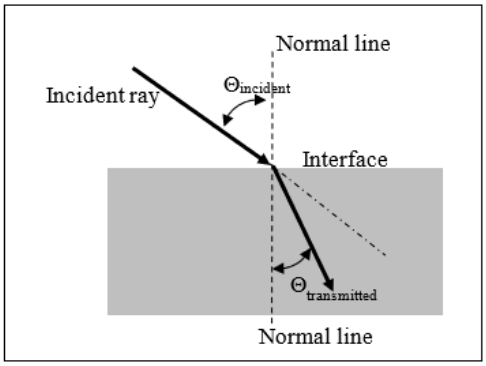
4) For EM waves traveling from in the opposite direction, the wave would have a lower velocity and then speed up in the new material, this speeding up would bend the light in the other direction and increase the wavelength.

From the wave equation

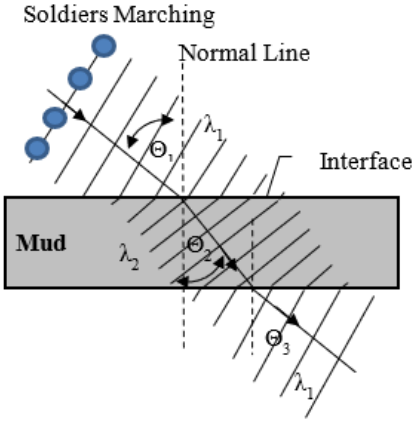
$$v_{material} = f\lambda_{\text{material}} \quad and \quad f = \frac{v_{material}}{\lambda_{material}}$$

$$\lambda_{\text{material}} = \frac{v_{material}}{f}$$

- When the velocity of the wave is reduced the wavelength must get smaller to keep the frequency constant.



Example: If there are Soldiers marching in a straight line then as they march some of the line starts to march in mud. The soldiers that are matching in the mud will slow down compared to the others in the grass; the result will be a bending of the line of soldiers.



If the soldiers are marching in a line in the grass at 2 m/s with a frequency of 1 Hz (one soldiers per second that go by a point), what would their wavelength (separation between the lines of soldiers) in the grass be and when they hit the mud what would their wavelength be if they slowed down to 1 m/s.

$\lambda_{grass} = v_{grass}/f = (2\text{ m/s})/(1\text{ Hz}) = 2\text{ m}$

$\lambda_{mud} = v_{mud}/f = (1\text{ m/s})/(1\text{ Hz}) = 1\text{ m}$

Index of refraction (n) (or refractive index) – is the ratio of the speed of light in vacuum to the speed of light in the material. The Index of Refraction indicates the amount speed of the EM wave is reduced due to the medium it is traveling through.

$$\mathbf{n = \frac{c}{v_{material}}}$$

n – index of refraction of the material
c – speed of light in vacuum = 3×10^8 m/s
$v_{material}$ – velocity of light in the material

- It is an indication of the amount that light is slowed down/bent.
- The larger the Index of Refraction the less speed and the more the light is bent.
- The smaller the index of refraction the less of a slowdown and the less the light is bent.

- Index of refraction must be greater than 1.0 because v is always less than or equal to the speed of light in vacuum (c).

$$v_{material} = \frac{c}{n}$$

- The frequency of the EM wave does not change when the index of refraction changes, but the wavelength of the EM wave changes by a factor of one over the index of refraction.

Index of Refraction (n)

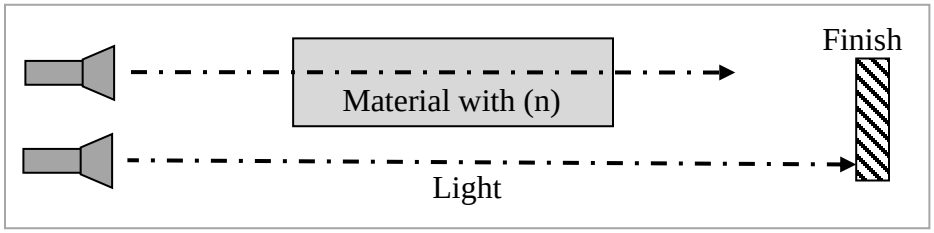
	n	velocity of Light
Vacuum/Air	1.00	3.0×10^8 m/s
Water	1.33	2.6×10^8 m/s
Glass	1.52	2.0×10^8 m/s
Diamond	2.42	1.2×10^8 m/s
(n measured at λ = 589 nm)		

Since the frequency of the light is constant $f_{material} = f_{vacuum}$, the wavelength ($\lambda_{material}$) of the EM wave in the material is reduced by a factor of the index of refraction.

$$\lambda_{material} = \frac{\lambda_{vaccum}}{n}$$

- The index of refraction can be thought of as converting the wave properties of the EM wave in material "A" to the wave properties of material "B".
- The index of refraction depends on the temperature of material and the frequency of light.

Example: If two flashlights were turned on at exactly the same time, the top path had a piece of glass and the bottom path did not, which light would reach the finish first? The light on the bottom would not be slowed down by the block of glass so it would reach the finish first.



Example: Green light with a wavelength of 530 nm enters a block of material with a length of 1 m and the index of refraction of n = 2.0.

A) How fast is the light traveling in the material and how long does it take to go the 1 meter? For the speed of light inside the material:

$v = c/n = (3 \times 10^8 \text{ m/s})/(2) = 1.5 \times 10^8 \text{ m/s}$

$v = \Delta x/\Delta t$ therefore $\Delta t = \Delta x/v = (1 \text{ m})/(1.5 \times 10^8 \text{ m/s}) = 6.7 \times 10^{-9} \text{ s} = 6.7 \text{ ns}$

B) How long does it take light to travel 1 m in air and how does this compare to the block above?

$v = \Delta x/\Delta t$ therefore $\Delta t = \Delta x/v = (1 \text{ m})/(3 \times 10^8 \text{ m/s}) = 3.3 \times 10^{-9} \text{ s} = 3.3 \text{ ns}$

The light in vacuum would reach the finish twice as fast.

C) What is the wavelength and frequency of the green light (λ_{air} = 530 nm) inside the material? For the frequency $v = f\lambda = c$ in air

$f = c/\lambda = (3 \times 10^8 \text{ m/s})/(530 \times 10^{-9} \text{ m}) = 5.66 \times 10^{14} \text{ Hz}$

the frequency of the light is the same for air or in the material

For the wavelength $v = f\lambda$, where v is the velocity of the light in the material

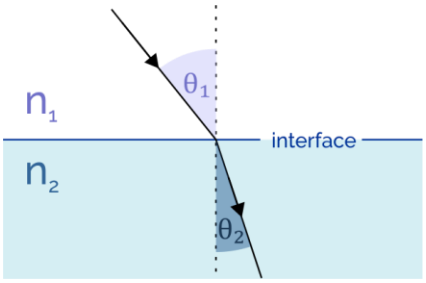
$\lambda_{material} = v_{material}/f = (1.5 \times 10^8 \text{ m/s})/(5.66 \times 10^{14} \text{ Hz}) = 2.65 \times 10^{-7} \text{ m} = 265 \text{ nm}$

Snell's Law (or Descartes Law) - Snell's Law is a mathematical description of the bending of light at the interface of two different material with different index of refraction.

$$n_1 \sin\Theta_1 = n_2 \sin\Theta_2$$

n_1 – index of material where the light is coming from
n_2 – index of material where the light is going into

Θ – measured from the normal (perpendicular) to the interface.
$\Theta_i = \Theta_1$ – indecent light ray in material n_1.
$\Theta_r = \Theta_2$ – refracted light ray in material n_2.



<u>For calculating the refracted angle</u>

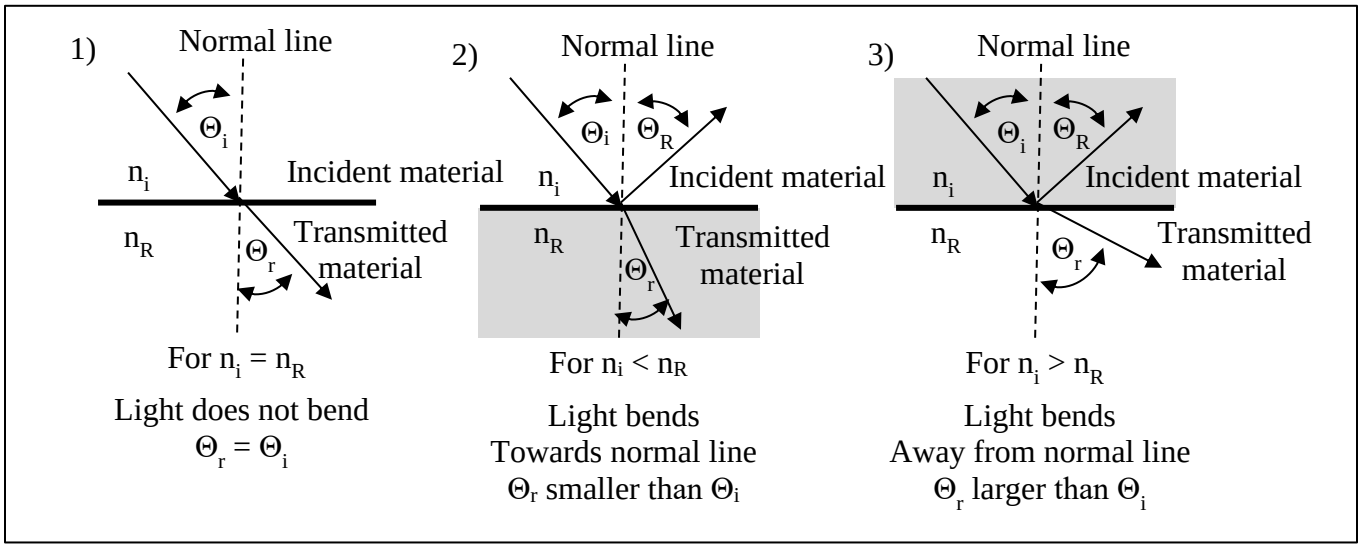
$$\Theta_2 = \sin^{-1}\left[\frac{n_1 \sin\Theta_1}{n_2}\right]$$

- At $\Theta_i = 0°$ the light will slow down but does not bend instead goes in a straight line.

Light travels from the incident material into the transmitted material.

- ✓ If $n_1 < n_2$ light is bent toward normal
- ✓ If $n_1 > n_2$ light is bent away from normal

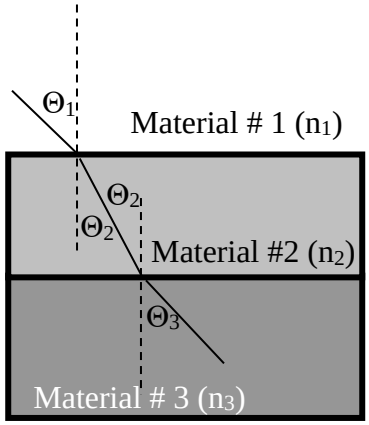
Given three interfaces below 1) no index of refraction change, 2) incident index of refraction smaller than transmitted index of refraction and 3) incident index of refraction larger than transmitted index of refraction.



1) If the incident material has the same index of refraction as the transmitted material ($n_i = n_r$) then there is no bending of light ($\Theta_i = \Theta_r$) and there is no reflected light because there is no index of refraction change.
2) If the incident material has a lower index of refraction as the transmitted material ($n_i < n_r$) then the light is bent towards the normal line in the transmitted material ($\Theta_i > \Theta_r$) and there is reflected light because there is an index of refraction change.
3) If the incident material has a higher index of refraction as the transmitted material ($n_i > n_r$) then the light is bent away from the normal line in the transmitted material ($\Theta_i < \Theta_r$) and there is reflected light because there is an index of refraction change.

Example: What can be said about the relative magnitudes of n_1, n_2 and n_3?

- Because Θ_2 is smaller than Θ_1, the light is refracted towards the normal, the light must be going from lower index to higher index therefore $n_1 < n_2$.
- Because Θ_3 is larger than Θ_2, the light is refracted away from the normal, the light must be going from higher index to lower index therefore $n_2 > n_3$.
- Because Θ_3 is slightly smaller than Θ_1, the light is still refracting towards the normal, then $n_1 < n_3$.



Therefore

$$n_1 < n_3 < n_2$$

Example: Using the diagram above assume Material # 1 is air with an index of refraction $n_{air} = 1.0$ and the light is incident on material # 2 at an angle of $\Theta_1 = 50°$, assume Material # 2 is glass with an index of refraction of 1.55.

A) What is the refracted angle in the glass?

$$n_{air}\sin\Theta_{air} = n_{glass}\sin\Theta_{glass}$$

$$\Theta_{glass} = \sin^{-1}\left[\frac{n_{air}\sin\Theta_{air}}{n_{glass}}\right] = 29.6°$$

✓ Going from low index of refraction to higher index of refraction the refracted angle is smaller (Incident angle of 50° to refracted angle of 29.6°)

B) If material # 3 is water with an index of refraction of 1.33, what is the refracted angle in the water from the light coming from the glass?

$$n_{glass}\sin\Theta_{glass} = n_{water}\sin\Theta_{water}$$

$$\Theta_{water} = \sin^{-1}\left[\frac{n_{glass}\sin\Theta_{glass}}{n_{water}}\right] = 35.14°$$

✓ Going from a high index of refraction to lower index of refraction the refracted angle is larger (Incident angle of 29.6° to refracted angle of 35.14°)

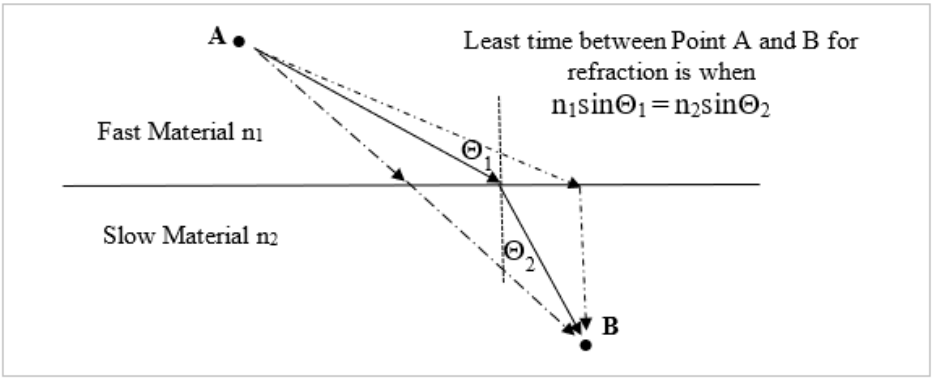
C) If the material on the other side of the water was air what would the refracted angle out of the water and back into the air?

$$\Theta_{air} = \sin^{-1}\left[\frac{n_{water}\sin\Theta_{water}}{n_{air}}\right] = 50°$$

- ✓ Going from a high index of refraction to lower index of refraction the refracted angle is larger (Incident angle of 35.14° to refracted angle of 50°)

Fermat's Principle for Refraction – Light takes the path of least time (not least distance). Both the Law of Reflection and Snell's Law can be derived from Fermat's Principle

Refraction – The path that takes the least amount of time to reach point B from point A is at an angle to the interface; it is not straight path or a perpendicular path.



Light and the Index of Refraction applications

Fresnel Reflection – Reflection of light due to the change in index of refraction. Its magnitude is dependent on the relative change in the index of refraction, the angle of incident and polarization of the light. Where R is the reflection coefficient.

$$R = \left[\frac{(n_1 - n_2)}{(n_1 + n_2)}\right]^2$$



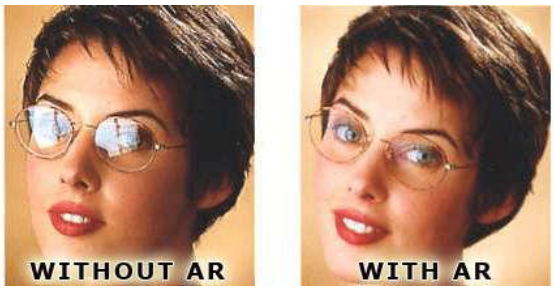
- This equation only applies to light rays normal to surface ($\Theta = 0°$) relative to the normal to the surface.

Example: For glass to air interface $n_{air} = 1.0$ and $n_{glass} = 1.52$ reflection what percentage of the light is reflected?

$$R = \left[\frac{(n_1 - n_2)}{(n_1 + n_2)}\right]^2 = \left[\frac{(1.0 - 1.52)}{(1.0 + 1.52)}\right]^2 = 0.043 \; or \; 4.3\,\%$$

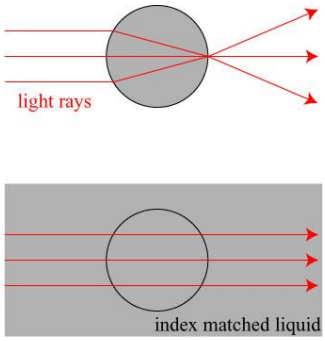
- For an air glass interface, the about of light reflected off the glass is about 4 %.
- The cat sees its reflection in the glass due to the change in the index of refraction.

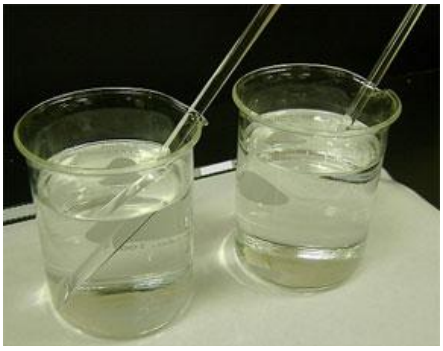
Anti-reflection (AR) coatings – AR coating can nearly eliminate Fresnel Reflection by using the interference of light waves. The reflected part of the wave is eliminated due to destructive interference; all the light is transmitted and therefore you do not "see" the surface



Index Matching - When the index of refraction of an object is the same as the fluid that it is submerged in, then there is no reflection at the interface between the object and fluid therefore there is no light reflected back from the surface in order to see the object. There is no distortion of the light rays due to bending of the light going through the object (see image below), the object in the index matching fluid appears invisible.

The first beaker has a glass rod in water, water and glass have different index of refractions (n_{glass} = 1.47 and n_{water} = 1.33) therefore the rod can be seen in the water. The second beaker has the same glass rod but with oil, oil and glass have the same index of refraction (n_{glass} = 1.47 and n_{oil} = 1.47) therefore the light passes through undistorted and there is no reflection back off the rod's surface.



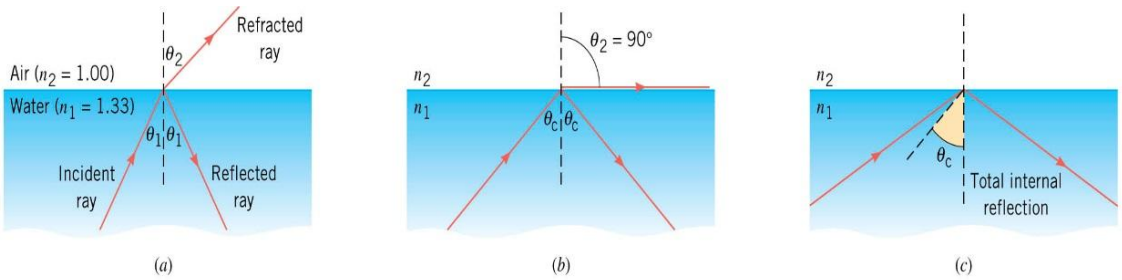


Total Internal Reflection – When the angle of refraction exceeds 90° (Θ_2) in the medium, all light is reflected, no refraction. There is no light transmitted through the interface for angles larger than the critical angle.

This effect only happens when going from a material of higher index of refraction to a material of lower index of refraction. The minimum angle to achieve total internal reflection is called the critical angle (Θ_c)

$$\mathrm{Sin}\Theta_c = n_2/n_1 = n_t/n_i \quad (n_2 < n_1)$$

$$\Theta_C = \sin^{-1}\left(\frac{n_{low}}{n_{high}}\right)$$



- There is no Total Internal Refection for $n_2 > n_1$, Why? Answer: the refracted angle will never equal to 90°.

Example: If Cathy is at the bottom of a pool filled with fresh water (n_{water} = 1.33) and looks up at the surface what angle relative to the normal does Cathy stop seeing through the water but only sees the reflection of the bottom of the pool?

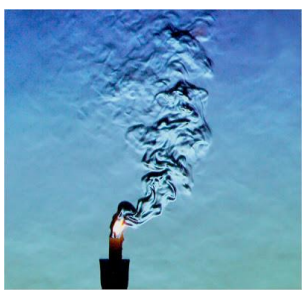
$$Sin\Theta_c = n_{air}/n_{water}$$

$$\Theta_c = Sin^{-1}(n_{aor}/n_{water}) = 48.7^o$$

- Any angle larger than this Cathy sees just reflection and any angle smaller than this and Cathy can see out the surface of the water.

Index of refraction and temperature – The index of refraction depends on the temperature of the material; this is especially true for gasses and liquids. For air the warmer the air the lower the index of refraction, the cooler the air the higher the index of refraction.

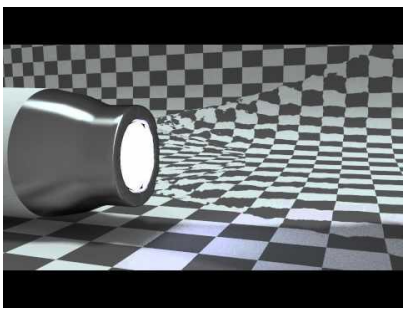
- The first image shows a candle, the hot and cool air causing change index of refration. The light passing through the hot and cool air is bent in many directions causing the light and dark areas.
- The second image shows the heat coming off a train track, the difference in index of refaction of the hot and cool air is show as light and dark shading and a image that is burred or hazy.
- The third image shows a hot air gun blowing hot air across an image of black and white squares, the squares are distored because of the bending of light due to the change in index of refraction of the hot and cooler air.



Candle



Haze



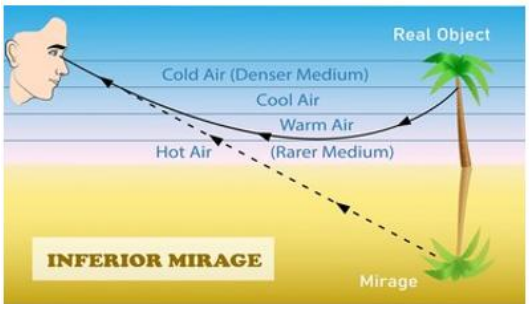
Heat gun

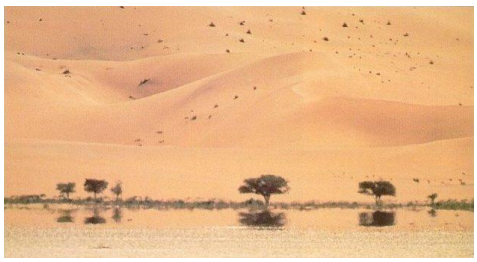
Mirages – Bending of light due to the different index of refractions of cold and warm air interface. There are two types of mirages depending on the orientation of the warm and cool air interface.

Inferior mirage – When cold air is above a thin layer of warm air in the environment, and the light from an object that is directed downward is bent upward due to the change in the index of refraction of the air (cool/hot interface). The observer "sees" the object in the ground because the observer assumes the light rays are straight from the object not bent. To an observer stand some distance away from the object, an inverted image (the bent light) of the object appears under the object. This effect can be seen on hot roads and in deserts. It is most famously seen as a "puddle of water" on a hot road or as an "oasis of water" in a desert.

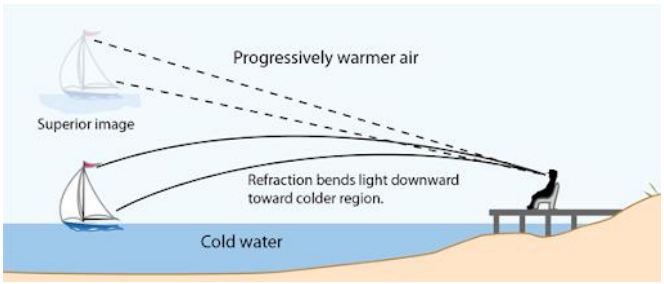
Superior Mirage When warm air is above a thin layer of cold air, the light rays are bent downward from the object (ship) due to the change in refractive index, so the image appears above the ship. The light is refracted downward, not upward as in an Inferior Mirage.

Inferior Mirage



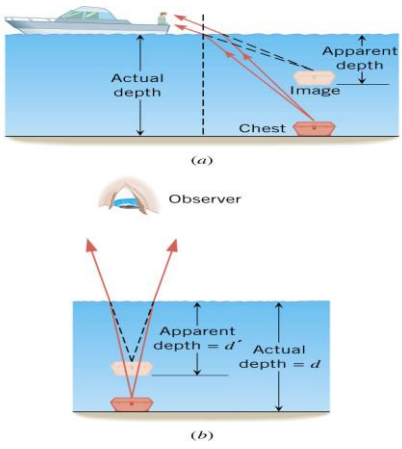


Superior Mirage





Apparent depth - A submerged object appears to a person looking down through the fluid as not being as deep as it actually is, the apparent depth is smaller than the actual depth of the object due to the refraction of light as it travels from the water into the air. Where d' is the apparent depth and d is the actual depth, n_1 is the index of the fluid the object is in (usually water) and n_2 is the index that the object is viewed from (usually air).



$$\frac{Apparent\ depth}{True\ depth} = \frac{d'}{d} = \frac{n_{air}}{n_{fluid}}$$

$$d' = d\left[\frac{n_1}{n_2}\right] = d\left[\frac{n_{air}}{n_{water}}\right]$$

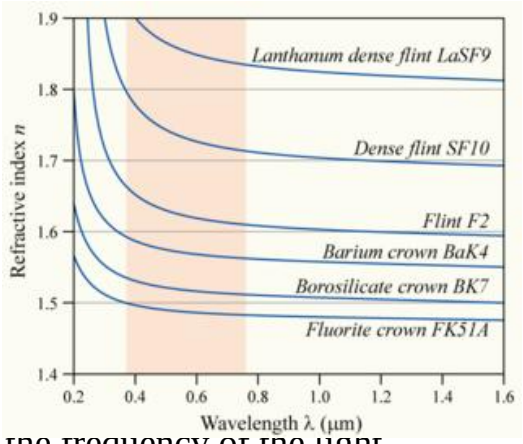
Example: Bob is looking down into a swimming pool ($n_1 = n_{water} = 1.33$) from air ($n_2 = n_{air} = 1.0$), he sees a dime at the bottom of the pool which is 3 meters deep, how deep does the dime appear to be to Bob?

$$d' = d\left[\frac{n_1}{n_2}\right] = (3\text{ m})\left[\frac{1.0}{1.33}\right] = 2.26\,m$$

Dispersive Media – Material for which light of different frequencies travels at different speeds. In Dispersive material the index of refraction depends on the frequency of the light.

In general, for the same dispersive material

- Higher frequencies have higher index of refraction
- Lower frequencies have a smaller index of refraction.
- In the visible spectrum blue has a higher index so it is bent more.
- Red light has a smaller index, so it is bent less.

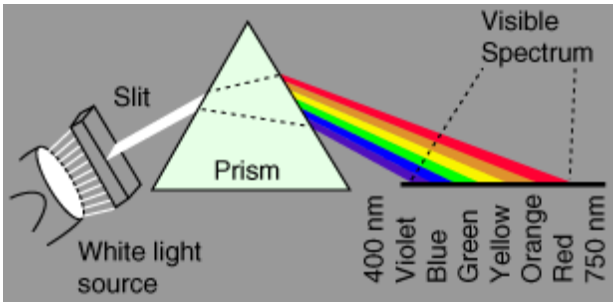


n(f) is the index of refraction that is a function of the frequency of the light.

$$v(f) = \frac{c}{n(f)}$$

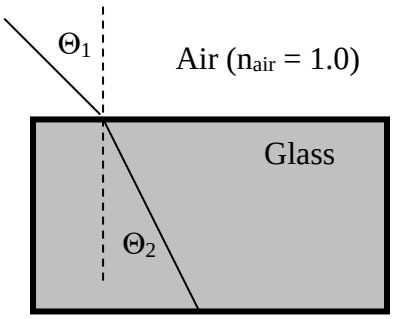
Prism – A prism separates white light into the color spectrum due to the different index of refraction of the different colors (frequencies of light). The separation of colors is due to the dispersion of the light in the prism material.

- The red light has a lower index of refraction therefore travels faster in the prims and is bent less.
- The blue light has a higher index of refraction therefore travels slower in the prism and is bent more.



- If the material has parallel surfaces, then there is no dispersion because the first wave to enter material is the first wave to leave material.
- For the same glass n(red) = 1.520 and n(violet) = 1.538.

Example: If a ray of light ($\Theta_1 = 60°$) goes from air into a block of glass, what would be the refracted angle for the red light and the blue light?



$n_{red} = 1.52$
$n_{blue} = 1.54$

$$\Theta_{red} = \sin^{-1}\left[\frac{n_{air}\sin\Theta_{air}}{n_{glass}}\right] = 34.73°$$

$$\Theta_{blue} = \sin^{-1}\left[\frac{n_{air}\sin\Theta_{air}}{n_{glass}}\right] = 34.22°$$

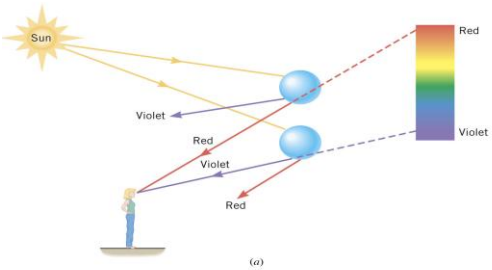
✓ The blue light is refracted more than the red

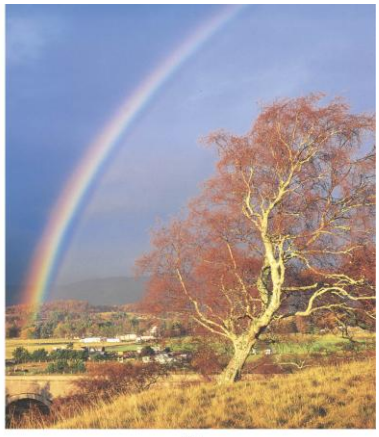
Rainbows – Rainbows are formed when there are small spherical water droplets in the air (rain) and the suns light enters the droplet and is refracted and total internally reflected off the back surface of the water droplets. The refraction is different for different colors (dispersion) so the white light is separated into its colors. This effect can only be seen when the water droplets are in front of the viewer and the sun is behind. Also, the light comes out of the droplets at a fixed angle (between 40° – 42°) therefore, as you move closer to the cloud the rainbow will not be visible.

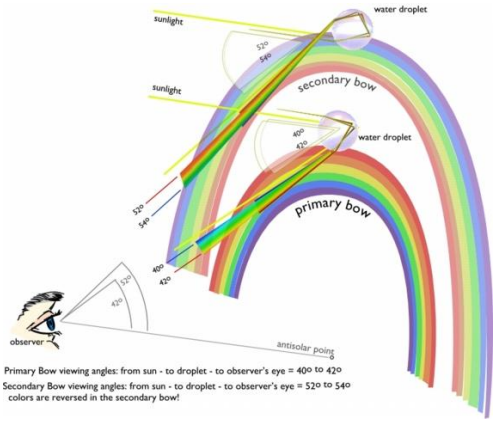
- The rainbow would be a complete circle if not for the earth blocking the other side.

One can never reach the "end of the rainbow" because in order to see the rainbow you have to be at the correct angle, the closer one get to the water droplets the rainbow will vanish.

❖ A double rainbow is formed when the light bounces a second time in the water droplet, so it comes out at a slightly different angle.



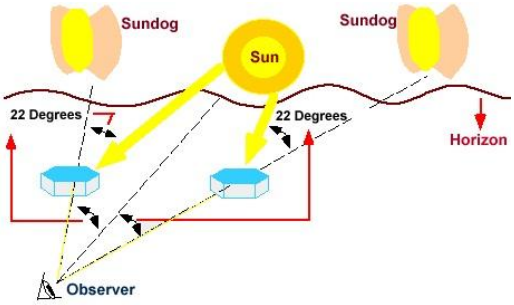




Sun dogs – are rainbow like colors that appear in high altitude clouds due to the alignment of the hexagonal ice crystals aligning so that the light from the sun is refracted to the viewer on the ground. The sun dog is part of a 360° halo that is formed around the sun due to the ice crystals.





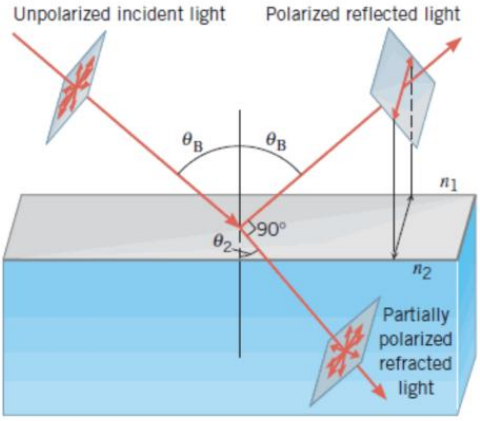


Brewster's Angle - Polarization of light and the index of refraction - Is the angle at which the reflected light is completely polarized parallel to the surface and the refracted light only being partially polarized.

$$\mathrm{Tan}(\theta_B) = \frac{n_{material}}{n_{incident}} \qquad \Theta_B = Tan^{-1}\left(\frac{n_2}{n_1}\right)$$

where n_1 is usually air $n_{air} = 1.0$

- The angle between reflected and refracted ray is equal to 90º.



Example: For light reflecting off of water (n = 1.33) and air (n = 1.0) determine Brewster's angle for the total polarization of the reflected light.

$\Theta_B = \mathrm{Tan}^{-1}(n_{water}/n_{air}) = \mathrm{Tan}^{-1}(1.33/1.0) = 53°$ or 37° from the surface of the water

- The reflected light of the clouds in the water are near Bewsters angle, when a polarizer with its axis of polarization vertial is introduced the reflected light is eliminted.



No Polarizer With Polarizer